

ISTARVSHAN, Tajikistan

WMO: 387130

Lat: 39.900N

Lon: 68.983E

Elev: 3297

StdP: 13.03

Time Zone: 5.00 (E05)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.3	(c) 15.4	(d) 1.6	(e) 6.7	(f) 12.6	(g) 5.6	(h) 8.3	(i) 18.7	(j) 10.0	(k) 37.6	(l) 9.5	(m) 36.0	(n) 3.8	(o) 140	(p) 0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.5	(c) 93.8	(d) 64.4	(e) 91.4	(f) 63.0	(g) 88.9	(h) 62.4	(i) 66.7	(j) 89.6	(k) 65.2	(l) 87.0	(m) 64.0	(n) 84.7	(o) 4.9	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.4	82.5	73.4	57.0	78.2	73.3	55.5	74.2	73.2	33.1	90.1	31.9	87.1	30.9	85.0	70.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.7	(b) 9.2	(c) 7.7	(e) 7.1	(f) 97.9	(g) 6.0	(h) 3.0	(i) 2.8	(j) 100.1	(k) -0.8	(l) 101.8	(m) -4.2	(n) 103.5	(o) -8.5	(p) 105.7		
(4) 10.7	(5) 9.2	(6) 7.7	(7) 7.1	(8) 97.9	(9) 6.0	(10) 3.0	(11) 2.8	(12) 100.1	(13) -0.8	(14) 101.8	(15) -4.2	(16) 103.5	(17) -8.5	(18) 105.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	32.0	33.7	43.4	53.0	63.5	73.1	77.0	73.8	65.4	53.1	43.0	35.3	(6)
(7)		DBStd	17.44	7.30	9.30	8.73	7.80	6.66	5.21	4.65	5.06	6.54	8.73	9.04	8.66	(7)
(8)		HDD50	2091	557	459	241	55	7	0	0	0	3	71	241	457	(8)
(9)		HDD65	5072	1022	875	671	364	105	5	0	2	70	377	662	919	(9)
(10)		CDD50	3541	0	4	35	145	425	692	838	738	466	166	29	3	(10)
(11)		CDD65	1048	0	0	0	4	58	247	373	275	83	7	1	0	(11)
(12)		CDH74	11472	0	0	1	53	564	2591	4331	2900	939	86	7	0	(12)
(13)		CDH80	4633	0	0	0	5	117	1013	2020	1198	266	13	1	0	(13)
(14)	Wind	WSAvg	4.1	3.9	3.9	4.5	4.5	4.6	4.6	4.4	4.2	3.9	3.9	3.6	3.6	(14)
(15)	Precipitation	PrecAvg	20.4	2.1	2.8	3.3	3.1	2.4	1.0	0.6	0.3	0.3	0.9	1.7	2.0	(15)
(16)		PrecMax	29.0	3.8	4.8	5.3	7.4	4.7	2.1	1.5	0.8	0.7	2.0	4.4	3.9	(16)
(17)		PrecMin	12.8	0.9	0.9	0.9	1.0	0.4	0.4	0.2	0.1	0.0	0.2	0.5	0.3	(17)
(18)		PrecStd	4.0	0.8	0.9	1.1	1.2	1.1	0.4	0.3	0.2	0.2	0.4	1.0	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	62.3	70.1	80.5	87.1	94.3	98.1	94.7	89.0	81.8	71.1	60.8	(19)
MCWB			42.0	47.5	52.7	60.2	62.0	63.4	66.4	65.2	59.7	58.6	51.2	44.9	(20)	
2%		DB	49.7	55.0	65.4	74.4	83.3	91.3	95.4	92.4	86.1	75.8	64.7	55.6	(21)	
		MCWB	39.0	43.5	50.9	56.4	61.3	62.0	65.0	64.1	59.6	55.0	48.3	43.7	(22)	
5%		DB	45.8	49.9	61.3	70.2	80.0	88.9	92.4	89.6	83.3	71.6	60.2	51.2	(23)	
		MCWB	37.3	40.8	49.9	55.1	60.2	62.1	64.1	61.9	59.3	53.2	46.7	41.1	(24)	
10%		DB	42.0	46.3	56.6	66.2	76.8	86.2	89.8	86.8	79.9	67.8	56.2	47.2	(25)	
		MCWB	35.1	38.9	46.6	54.0	59.5	61.4	63.4	61.6	58.1	51.4	45.2	39.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.7	48.0	54.7	60.8	64.6	66.6	68.7	68.0	62.3	58.9	53.5	47.2	(27)
MADB			52.8	60.7	66.4	74.4	80.7	88.0	93.5	90.9	83.5	80.5	66.0	57.7	(28)	
2%		WB	40.1	44.7	52.2	58.5	63.0	64.9	67.3	65.7	60.5	56.0	50.2	44.2	(29)	
		MADB	48.1	52.9	63.3	71.7	78.1	85.3	91.2	88.7	81.8	73.2	60.5	53.6	(30)	
5%		WB	37.8	41.9	50.1	56.4	61.6	63.8	65.9	64.1	59.5	54.1	48.2	42.0	(31)	
		MADB	44.7	48.9	60.0	67.6	75.9	83.5	88.8	86.2	81.2	69.8	58.4	49.1	(32)	
10%		WB	35.7	39.3	47.6	54.6	60.3	62.7	64.6	62.4	58.6	52.1	46.0	39.8	(33)	
		MADB	41.1	45.5	55.8	64.4	74.1	81.5	86.5	83.3	79.6	66.1	55.2	46.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	15.2	14.8	15.3	16.8	18.7	21.0	21.5	22.3	22.6	20.3	16.9	15.1	(35)	
MCDBR			19.1	18.9	21.2	22.4	23.5	24.3	24.6	24.5	26.6	24.9	23.1	19.9	(36)	
MCWBR			12.9	13.0	12.7	12.2	11.3	9.8	10.5	10.1	12.0	12.8	12.3	12.1	(37)	
5% WB		MCDBR	17.1	17.6	19.1	19.8	20.7	22.0	22.9	23.5	25.1	23.2	20.2	17.5	(38)	
		MCWBR	12.0	12.7	12.2	11.9	11.3	10.1	10.8	11.0	12.1	12.6	11.8	11.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.341	0.388	0.454	0.474	0.474	0.474	0.464	0.437	0.418	0.446	0.404	0.343	(40)	
taud		2.177	2.053	1.929	1.929	1.983	1.988	1.988	2.015	2.080	2.074	1.939	2.016	2.154	(41)	
Ebn at Noon		252	255	250	255	258	258	259	263	259	230	222	239	(42)		
Edh at Noon		33	43	55	58	56	56	54	49	47	48	39	31	(43)		
(44)	All-Sky Solar Radiation	RadAvg	634	876	1209	1592	1987	2322	2357	2163	1778	1174	709	550	(44)	
(45)		RadStd	64	114	124	186	140	102	99	107	65	116	98	73	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	+167	+121

Nomenclature: See separate page